Lecture 3: Metric Spaces and Topology

Mathematical Analysis

1 Metric Spaces

Definition 1.1 (Metric). [def:metric] Let X be a set. A *metric*, or *distance function*, on X is a function $d : X \times X \to \mathbb{R}$ satisfying the following four axioms: nonnegativity, identity of indiscernibles, symmetry, and the triangle inequality

Axiom 1.1 (Nonnegativity). [axm:metric-nonnegativity] *If d is a metric on X, then*

$$\forall x, y \in X, \quad d(x, y) \geq 0$$

Axiom 1.2 (Identity of Indiscernibles). [axm:metric-separation] *If d is a metric on X, then*

$$\forall x, y \in X, \quad d(x, y) = 0 \iff x = y$$

Axiom 1.3 (Symmetry). [axm:metric-symmetry] *If d is a metric on X, then*

$$\forall x, y \in X, \quad d(x, y) = d(y, x)$$

Axiom 1.4 (Triangle Inequality). [axm:metric-triangle-inequality] *If d is a metric on X, then*

$$\forall x, y, z \in X, \quad d(x, z) \leq d(x, y) + d(y, z)$$

Definition 1.2 (Metric Space). [def:metric-space] A *metric space* is a pair (X, d), where X is a set and d is a metric on X

2 Open Balls and Neighbourhoods

Definition 2.1 (Open Balls). [def:open-balls] Let (X, d) be a metric space. The *open-ball generator* is

$$\text{OpenBall} : \mathbb{R}_+ \to \big(X \to \mathcal{P}(X)\big) \equiv r \mapsto x \mapsto \{y \in X : d(x, y) < r\}$$

For $r \in \mathbb{R}_+$ and $x \in X$, $\text{OpenBall}(r)(x)$ is the *open ball centred at x with radius r*

`LaTeX: \OpenBall (r)(x)`

Remark 2.1 (Open Ball Intuition). [rem:open-ball-intuition] An open ball in a metric space (X, d), centred at $x \in X$ with positive radius r, is the collection of all elements of X whose distance from x is less than r

Convention 2.1 (Open Ball Notation). [conv:open-ball-notation] For the open-ball generator, write

$$B_r(x) \equiv \text{OpenBall}(r)(x)$$

`LaTeX: \ball {r}{x}`

Definition 2.2 (Neighbourhood). [def:neighbourhood] Let (X, d) be a metric space, and let $x \in X$. A set $U \subseteq X$ is a *neighbourhood of x* if

$$\exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U$$

Definition 2.3 (Bounded Set in a Metric Space). [def:metric-bounded-set] Let (X, d) be a metric space. A subset $S \subseteq X$ is *bounded* if

$$\exists M \in \mathbb{R}_+, \forall x, y \in S, \quad d(x, y) < M$$

Convention 2.2 (Radius Neighbourhood Notation). [conv:radius-neighbourhood-notation] In a metric space (X, d), some texts call the open ball $B_r(x)$, with $r \in \mathbb{R}_+$ and $x \in X$, the *r-neighbourhood of* x and write

$$V_r(x) \equiv B_r(x)$$

Lemma 2.1 (The Empty Set and the Whole Space). [lem:metric-empty-and-whole-open] *Let (X, d) be a metric space. Then*

$$\forall U \in \{\emptyset, X\},\ \forall x \in U,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U$$

Proof. **Case $U = X$:** For every $x \in X$, choose $r = 1$. By definition, $B_1(x) \subseteq X$
Case $U = \emptyset$: There is no $x \in \emptyset$, so the universally quantified condition holds vacuously
The empty-set case does not require finding a ball contained in $\emptyset$, since there is no centre to consider □

Lemma 2.2 (Finite Intersections). [lem:metric-finite-intersections] *Let (X, d) be a metric space, and let $U, V \subseteq X$ satisfy*

$$\forall x \in U,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U$$
$$\forall x \in V,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq V$$

Then

$$\forall x \in U \cap V,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U \cap V$$

Proof. Let $x \in U \cap V$. By the assumptions, choose $r_U, r_V \in \mathbb{R}_+$ such that $B_{r_U}(x) \subseteq U$ and $B_{r_V}(x) \subseteq V$. Let $r = \min(r_U, r_V) > 0$. Then

$$B_r(x) \subseteq B_{r_U}(x) \subseteq U, \qquad B_r(x) \subseteq B_{r_V}(x) \subseteq V$$

Therefore, $B_r(x) \subseteq U \cap V$ □

Lemma 2.3 (Arbitrary Unions). [lem:metric-arbitrary-unions] *Let (X, d) be a metric space, and let $U : I \to \mathcal{P}(X)$ satisfy*

$$\forall i \in I,\ \forall x \in U(i),\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq U(i)$$

Then

$$\forall x \in \bigcup_I U,\ \exists r \in \mathbb{R}_+, \quad B_r(x) \subseteq \bigcup_I U$$

Proof. Let $x \in \bigcup_I U$. Choose $i \in I$ with $x \in U(i)$. By the assumption, choose $r \in \mathbb{R}_+$ such that

$$B_r(x) \subseteq U(i) \subseteq \bigcup_I U$$

□

3 Distance on an Ordered Field

Definition 3.1 (Distance). [def:distance] Let F be an ordered field. For $x, y \in F$, the *distance* between x and y is

$$d(x, y) \equiv |x - y|$$

`LaTeX: \dist {x}{y}`

Lemma 3.1 (Symmetry of Distance). [lem:distance-symmetry] *Let F be an ordered field. For all $x, y \in F$,*

$$d(x, y) = d(y, x)$$

Proof. By definition, $d(x,y) = |x-y|$ and $d(y,x) = |y-x| = |-(x-y)|$. It suffices to show $|-(x-y)| = |x-y|$; we split on the sign of $x-y$
Case 1: $x-y \geq 0$. Then $|x-y| = x-y$. Also $-(x-y) \leq 0$, so $|-(x-y)| = -(-(x-y)) = x-y = |x-y|$
Case 2: $x-y < 0$. Then $|x-y| = -(x-y)$. Also $-(x-y) > 0$, so $|-(x-y)| = -(x-y) = |x-y|$ □

Lemma 3.2 (Triangle Inequality for Distance). [lem:triangle-inequality-distance] *Let F be an ordered field. For all $x, y, z \in F$,*

$$d(x,z) \leq d(x,y) + d(y,z)$$

Proof. Let $a = x-y$ and $b = y-z$. Then $x-z = a+b$, so $d(x,z) = |a+b|$. We split on the sign of $a+b$
Case 1: $a+b \geq 0$. Then $|a+b| = a+b$. Since $a \leq |a|$ and $b \leq |b|$,

$$|a+b| = a+b \leq |a| + |b| = d(x,y) + d(y,z)$$

Case 2: $a+b < 0$. Then $|a+b| = -(a+b) = -a-b$. Since $-a \leq |a|$ and $-b \leq |b|$,

$$|a+b| = -a-b \leq |a| + |b| = d(x,y) + d(y,z)$$

□

Corollary 3.1 (Metric on a Complete Ordered Field). [cor:complete-ordered-field-metric] *Let F be a complete ordered field, and let d be the absolute-value distance on F. Under the identification with the reals, (F,d) is a metric space*

Proof. Under this identification, the distance is real-valued. By the definition of absolute value,

$$d(x,y) = |x-y| \geq 0, \qquad d(x,y) = 0 \iff x-y = 0 \iff x = y$$

Symmetry and the triangle inequality give the remaining two metric axioms □

Lemma 3.3 (ε-Criterion for Equality). [lem:epsilon-equality] *Let F be an ordered field. For $a, b \in F$,*

$$a = b \iff \forall \varepsilon \in F_+,\ |a-b| < \varepsilon$$

Proof. In order to proof Epsilon Equality Lemma, we need to prove both directions
(a): $a = b \implies \forall \varepsilon \in F_+, |a-b| < \varepsilon$
Suppose $a = b$. Then $|a-b| = 0$. For any $\varepsilon > 0$, we have $|a-b| = 0 < \varepsilon$
(b): $\forall \varepsilon \in F_+, |a-b| < \varepsilon \implies a = b$
Assume $a \neq b$
Then, this means that

$$\exists \varepsilon_0 \in F_+, |a-b| = \varepsilon_0$$

Taking $\varepsilon = \varepsilon_0$ gives $\varepsilon_0 = |a-b| < \varepsilon_0$, a contradiction
So, we have $\forall \varepsilon \in F_+, |a-b| < \varepsilon \implies \neg a \neq b$
Since $\neg a \neq b \iff a = b$, So, $\forall \varepsilon \in F_+, |a-b| < \varepsilon \implies a = b$
Therefore, both implications hold, and hence proven □

4 Topological Spaces

Definition 4.1 (Topology)**.** [def:topology] Let X be a set. A *topology* on X, called $\mathcal{T}$, is a subset of $\mathcal{P}X$ satisfying containment of the empty set and the whole set, closure under finite intersections, and closure under arbitrary unions

Axiom 4.1. [axm:topology-empty-and-whole] *If $\mathcal{T}$ is a topology on X, then*

$$\emptyset, X \in \mathcal{T}$$

Axiom 4.2. [axm:topology-finite-intersections] *If $U, V \in \mathcal{T}$, then*

$$U \cap V \in \mathcal{T}$$

Axiom 4.3. [axm:topology-arbitrary-unions] *If $U : I \to \mathcal{T}$ is a family of sets in $\mathcal{T}$, then*

$$\bigcup_I U \in \mathcal{T}$$

Definition 4.2 (Topological Spaces)**.** [def:topological-space] A *topological space* is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T}$ is a topology on X

Definition 4.3 (Induced Topology)**.** [def:induced-topology] Let (X, d) be a metric space, where we define a *induced topology* $\mathcal{T}_d$ induced by d ;s

$$\mathcal{T}_d \equiv \{U \in \mathcal{P}(X) \mid \forall x \in U,\, \exists r \in \mathbb{R}_+,\, B_r(x) \subseteq U\}$$

Theorem 4.1 (Induced Topology is a Topology)**.** [thm:induced-topology-is-a-topology]

$$(X, d) \text{ is a metric space} \implies (X, \mathcal{T}_d) \text{ is a topological space}$$

Proof. By definition, $\mathcal{T}_d \subseteq \mathcal{P}(X)$. The lemmas on the empty set and the whole space, finite intersections, and arbitrary unions give

$$\emptyset, X \in \mathcal{T}_d, \qquad U, V \in \mathcal{T}_d \implies U \cap V \in \mathcal{T}_d, \qquad U : I \to \mathcal{T}_d \implies \bigcup_I U \in \mathcal{T}_d$$

Thus $\mathcal{T}_d$ satisfies the three topology axioms, so $(X, \mathcal{T}_d)$ is a topological space □

5 Open and Closed Sets

Definition 5.1 (Open Sets)**.** [def:open-sets] A set $U \subseteq X$ is *open* in a topological space $(X, \mathcal{T})$ if $U \in \mathcal{T}$.

Definition 5.2 (Closed Sets)**.** [def:closed-sets] A set $C \subseteq X$ is *closed* in a topological space $(X, \mathcal{T})$ if its complement $X \setminus C$ is open, i.e., $X \setminus C \in \mathcal{T}$.

Corollary 5.1 (Arbitrary Unions of Open Sets)**.** [cor:open-arbitrary-unions] *Let $(X, \mathcal{T})$ be a topological space, and let $U : I \to \mathcal{P}(X)$. If every $U(i)$ is open, then*

$$\bigcup(U) \in \mathcal{T}$$

Proof. Each $U(i) \in \mathcal{T}$ by the definition of an open set, so the arbitrary-union axiom gives $\bigcup(U) \in \mathcal{T}$ □

Corollary 5.2 (Finite Intersections of Open Sets). [cor:open-finite-intersections] *Let $(X, \mathcal{T})$ be a topological space, and let $U : I \to \mathcal{P}(X)$, where I is finite and nonempty. If every $U(i)$ is open, then*

$$\bigcap(U) \in \mathcal{T}$$

Proof. A single member belongs to $\mathcal{T}$. Applying the intersection axiom repeatedly proves the claim by induction on the number of sets □

Corollary 5.3 (Arbitrary Intersections of Closed Sets). [cor:closed-arbitrary-intersections] *In a topological space $(X, \mathcal{T})$, let $C : I \to \mathcal{P}(X)$, where I is nonempty. If every $C(i)$ is closed, then $\bigcap(C)$ is closed*

Proof. Each $X \setminus C(i)$ is open by the definition of a closed set. By De Morgan's laws and the open-union corollary,

$$X \setminus \bigcap(C) = \bigcup(i \mapsto X \setminus C(i)) \in \mathcal{T}$$

Thus $\bigcap(C)$ is closed □

Corollary 5.4 (Finite Unions of Closed Sets). [cor:closed-finite-unions] *Let $(X, \mathcal{T})$ be a topological space, and let $C : I \to \mathcal{P}(X)$, where I is finite. If every $C(i)$ is closed, then $\bigcup(C)$ is closed*

Proof. If I is empty, then $\bigcup(C) = \emptyset$ is closed since $X \in \mathcal{T}$. Otherwise, each $X \setminus C(i)$ is open. By De Morgan's laws and the open-intersection corollary,

$$X \setminus \bigcup(C) = \bigcap(i \mapsto X \setminus C(i)) \in \mathcal{T}$$

Thus $\bigcup(C)$ is closed □

Proposition 5.1 (Open Balls are Open). [prop:metric-open-balls-open] *In a metric space (X, d), every open ball is open in the induced topology*

Proof. Let $z \in B_r(x)$, and choose $s = r - d(x, z) > 0$. For $y \in B_s(z)$, the triangle inequality gives

$$d(x, y) \leq d(x, z) + d(z, y) < d(x, z) + s = r$$

Thus $B_s(z) \subseteq B_r(x)$, so $B_r(x)$ is open □

6 Limit Points and Isolated Points

Definition 6.1 (Limit Points). [def:limit-points] Let (X, d) be a metric space. A *limit point* $x \in X$ of a set $U \subseteq X$ is a point such that

$$\forall \varepsilon \in \mathbb{R}_+, \quad B_\varepsilon(x) \cap \big(U \setminus \{x\}\big) \neq \emptyset$$

Redirect (Sequence Characterization of Limit Points) [thm:limit-point-sequence]
Moved to theorem limit point sequence.

7 Compact Sets

Definition 7.1 (Cover). [def:cover] Let X be a set and $S \subseteq X$. A *cover* of S is a family $U : I \to \mathcal{P}(X)$ such that

$$S \subseteq \bigcup_I U$$

Definition 7.2 (Finite Cover). [def:finite-cover] A cover $U : I \to \mathcal{P}(X)$ of $S \subseteq X$ is *finite* if its index set I is finite, equivalently,

$$|I| < |\mathbb{N}|$$

Definition 7.3 (Open Cover). [def:open-cover] Let $(X, \mathcal{T})$ be a topological space, and let $S \subseteq X$. A cover $U : I \to \mathcal{P}(X)$ of S is an *open cover* if

$$\forall i \in I, \quad U(i) \in \mathcal{T}$$

Definition 7.4 (Subcover). [def:subcover] Let $U : I \to \mathcal{P}(X)$ be a cover of $S \subseteq X$. A *subcover* of U is a cover $V : J \to \mathcal{P}(X)$ of S such that

$$\forall j \in J, \exists i \in I, \quad V(j) = U(i)$$

The subcover is *finite* if V is a finite cover

Definition 7.5 (Compact Set). [def:compact-set] Let $(X, \mathcal{T})$ be a topological space. A subset $S \subseteq X$ is *compact* if every open cover of S has a finite subcover

8 The Usual Metric and Topology on $\mathbb{R}$

Definition 8.1 (Usual Topology on the Real Line). [def:usual-topology-real] Equip $\mathbb{R}$ with the absolute-value distance, which is a metric:

$$d(x, y) = |x - y|, \qquad x, y \in \mathbb{R}$$

The *usual topology* on $\mathbb{R}$ is the topology induced by the absolute-value metric d. In this section, balls and open sets in $\mathbb{R}$ refer to this metric and topology

Proposition 8.1 (Open Balls on the Real Line). [prop:real-open-balls] *For $x \in \mathbb{R}$ and $r \in \mathbb{R}_+$, the open ball is the open interval*

$$B_r(x) = (x - r, x + r)$$

Proof. For $y \in \mathbb{R}$,

$$y \in B_r(x) \iff |x - y| < r \iff x - r < y < x + r \iff y \in (x - r, x + r)$$

□

Corollary 8.1 (Open Intervals are Open). [cor:real-open-intervals] *For $a, b \in \mathbb{R}$ with $a < b$, the open interval (a, b) is open in the usual topology*

Proof. For each $x \in (a, b)$, we can choose a ball centred at x with radius $r = \min(x - a, b - x) > 0$. Then

$$B_r(x) = (x - r, x + r) \subseteq (a, b)$$

Thus every point of (a, b) has a ball contained in (a, b), so (a, b) is open by the definition of the induced topology □

Proposition 8.2 (Open Sets as Unions of Open Intervals). [prop:real-open-sets-intervals] *A set $U \subseteq \mathbb{R}$ is open in the usual topology if and only if it is a union of open intervals*

Proof. ($\Rightarrow$) Suppose U is open. Let

$$I \equiv \{(x, r) \in U \times \mathbb{R}_+ : B_r(x) \subseteq U\}, \qquad J : I \to \mathcal{P}(\mathbb{R}) \equiv (x, r) \mapsto (x - r, x + r)$$

Every $J(x, r) = B_r(x)$ is contained in U, and each $x \in U$ belongs to one such interval by the definition of $\mathcal{T}_d$. Hence $U = \bigcup(J)$
($\Leftarrow$) Each open interval is open, and arbitrary unions of open sets are open □

Proposition 8.3 (Closed Intervals are Closed). [prop:real-closed-intervals] *For $a, b \in \mathbb{R}$ with $a \leq b$, the closed interval $[a, b]$ is closed in the usual topology*

Proof. Let $x \in \mathbb{R} \setminus [a, b]$. If $x < a$, choose $r = a - x > 0$; if $x > b$, choose $r = x - b > 0$. In either case, the ball centred at x satisfies

$$B_r(x) = (x - r, x + r) \subseteq \mathbb{R} \setminus [a, b]$$

Thus $\mathbb{R} \setminus [a, b]$ is open, so $[a, b]$ is closed by the definition of a closed set □

9 Heine-Borel Theorem

Lemma 9.1 (Compact Sets are Bounded). [lem:compact-implies-bounded] *Let (X, d) be a metric space. Every compact subset $S \subseteq X$, in the induced topology, is bounded*

Proof. The empty set is bounded, so suppose $S \neq \emptyset$ and choose $x_0 \in S$. The family

$$U : \mathbb{R}_+ \to \mathcal{P}(X) \equiv r \mapsto B_r(x_0)$$

is an open cover of S: each ball is open, and each $x \in S$ belongs to $B_{d(x_0,x)+1}(x_0)$. Compactness gives a finite subcover. Let $R > 0$ be the largest of its finitely many radii; then $S \subseteq B_R(x_0)$. For all $x, y \in S$,

$$d(x, y) \leq d(x, x_0) + d(x_0, y) < 2 \cdot R$$

Hence S is bounded □

Lemma 9.2 (Compact Sets are Closed). [lem:compact-implies-closed] *Let (X, d) be a metric space. Every compact subset $S \subseteq X$, in the induced topology, is closed*

Proof. The empty set is closed, so suppose $S \neq \emptyset$. Fix $x \in X \setminus S$. For each $y \in S$, let $r_y = d(x, y)/2 > 0$. The balls $B_{r_y}(y)$ form an open cover of S, so finitely many, centred at $y_1, \ldots, y_k$, cover S. Let $r > 0$ be the smallest of their radii. If $z \in B_r(x) \cap B_{r_{y_j}}(y_j)$, then

$$d(x, y_j) \leq d(x, z) + d(z, y_j) < r + r_{y_j} \leq 2 \cdot r_{y_j} = d(x, y_j)$$

a contradiction. Thus $B_r(x) \subseteq X \setminus S$. Since x was arbitrary, $X \setminus S$ is open, so S is closed □

Lemma 9.3 (Closed and Bounded Sets are Compact). [lem:real-closed-bounded-compact] *Every subset $S \subseteq \mathbb{R}$ that is closed in the usual topology and bounded is compact*

Proof. The empty set is compact, so suppose $S \neq \emptyset$. Choose $a < b$ with $S \subseteq [a, b]$, and let U be any open cover of S. Add the open set $\mathbb{R} \setminus S$ to obtain an open cover $\mathcal{V}$ of $[a, b]$. Define

$$E \equiv \{t \in [a, b] : [a, t] \text{ has a finite subcover from } \mathcal{V}\}$$

Since one member covers a, we have $a \in E$, and b is an upper bound. By completeness, let $c = \sup(E)$
Choose V from $\mathcal{V}$ containing c, and choose $\delta > 0$ with $(c - \delta, c + \delta) \subseteq V$. By the supremum criterion, choose $t \in E$ with $c - \delta/2 < t \leq c$. A finite cover of $[a, t]$, together with V, covers

$$[a, \min(b, c + \delta/2)]$$

If $c < b$, this gives an element of E greater than c, contradicting its definition. Hence $c = b$, and the same finite collection covers $[a, b]$. Discarding $\mathbb{R} \setminus S$, if present, leaves a finite subcover of S from U. Thus S is compact □

Theorem 9.1 (Heine-Borel Theorem). [thm:heine-borel] *A subset $S \subseteq \mathbb{R}$ is compact in the usual topology if and only if it is closed and bounded*

Proof. A compact set is closed and metrically bounded. For a nonempty such set, fix $x_0 \in S$ and $M > 0$ with $|x - y| < M$ for all $x, y \in S$. Then $S \subseteq [x_0 - M, x_0 + M]$, so it is bounded in the order sense as well; the empty set is also bounded. The converse is the preceding lemma □

10 Cantor Sets

Definition 10.1 (Cantor Stages). [def:cantor-stages] The function $A : \mathbb{N} \to \mathcal{P}(\mathbb{R})$ is defined recursively by

$$A(0) \equiv [0, 1], \qquad A(n + 1) \equiv \left\{ \frac{x}{3} : x \in A(n) \right\} \cup \left\{ \frac{x + 2}{3} : x \in A(n) \right\}$$

Each stage removes the open middle third of every remaining interval, keeping its endpoints

Example 10.1 (The First Cantor Stages). [eg:cantor-first-stages]

$$A(1) = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1]$$
$$A(2) = [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{1}{3}] \cup [\tfrac{2}{3}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1]$$

Definition 10.2 (Middle-Third Cantor Set). [def:cantor-set] The *middle-third Cantor set* is the intersection of all Cantor stages:

$$\mathcal{C} \equiv \bigcap(A)$$

Thus $x \in \mathcal{C}$ if and only if x belongs to every $A(n)$

Proposition 10.1 (Properties of the Cantor Stages). [prop:cantor-stage-properties] *For each $n \in \mathbb{N}$, the stage $A(n)$ is a union of 2^n disjoint closed intervals of length 3^{-n}. Moreover,*

$$\{0, 1\} \subseteq A(n + 1) \subseteq A(n) \subseteq [0, 1]$$

Proof. The interval statement follows by induction: $A(0) = [0, 1]$, and the recurrence makes two copies of each interval, each scaled by $1/3$. The left copy lies in $[0, 1/3]$ and the right copy in $[2/3, 1]$, so the intervals remain disjoint. Also $A(1) \subseteq A(0)$, and the recurrence preserves inclusion, which proves nesting by induction. The points 0 and 1 belong to every stage, again by induction □

Corollary 10.1 (The Cantor Set is Compact). [cor:cantor-compact] *The Cantor set is a nonempty compact subset of $\mathbb{R}$*

Proof. Every $A(n)$ is closed, since closed intervals are closed and finite unions of closed sets are closed. Hence $\mathcal{C} = \bigcap(A)$ is closed by closure under intersections. Since $\{0, 1\} \subseteq \mathcal{C} \subseteq [0, 1]$, it is nonempty and bounded, so Heine–Borel gives compactness □